Weather decoder

Edition Oct 2026

A METAR, group by group (example)

METAR KCRQ 211455Z 00000KT 5SM BR FEW008 BKN015 14/13 A3001 RMK AO2 SLP163 T01390128

Code	Means
METAR KCRQ 211455Z	Routine report (SPECI special), station, 21st at 1455 UTC; AUTO automated, COR corrected
00000KT	Calm. Otherwise direction from, **true**, then knots; G = gust, VRB = variable
5SM BR	Visibility 5 statute miles in mist (5/8 to under 7 SM; FG below 5/8)
FEW008 BKN015	Few at 800 ft; broken at 1,500 ft AGL, the ceiling
14/13 A3001	Temperature/dewpoint °C (M = minus); altimeter 30.01 inHg
RMK AO2 SLP163 T01390128	Automated station with a precipitation discriminator; sea-level pressure 1016.3 mb; 13.9/12.8 °C

Also in remarks: PK WND peak wind · WSHFT wind shift (FROPA: frontal passage) · PRESRR / PRESFR pressure rising / falling rapidly.

Present weather

– light, **+** heavy, none moderate; **VC** 5–10 SM away

MI shallow **PR** partial **BC** patches **DR** low drifting **BL** blowing **SH** showers **TS** thunderstorm **FZ** freezing

DZ drizzle **RA** rain **SN** snow **SG** snow grains **IC** ice crystals **PL** ice pellets **GR** hail **GS** snow pellets **UP** unknown

BR mist **FG** fog **FU** smoke **VA** volcanic ash **DU** dust **SA** sand **HZ** haze **PY** spray

PO dust/sand whirls **SQ** squalls **FC** funnel cloud (+FC tornado or waterspout) **SS** sandstorm **DS** dust storm

Sky cover

SKC or **CLR** clear (CLR: automated, nothing at or below 12,000 ft)

FEW 1/8–2/8 · **SCT** 3/8–4/8

BKN 5/8–7/8 · **OVC** 8/8

VV vertical visibility into an obscuration

Ceiling: the lowest BKN or OVC layer, or VV. Heights in hundreds of feet AGL.

TAF change groups

FM211800	From the 21st, 1800Z: a complete new forecast replaces all before it
TEMPO 2112/2115	Temporary: over 50% likely, each 1 h or less, under half the period in all
PROB30 2123/2201	30% chance of thunderstorms or precipitation
WS020/27055KT	Wind shear up to 2,000 ft AGL; wind there 270° at 55 kt

P6SM = more than 6 SM. BECMG appears in military and international TAFs, not NWS TAFs.

Winds and temps aloft (FB), example

FT 3000 6000 12000 34000

SAN 0507 2611+12 2824-01 781550

0507: 050° at 7 kt, no temp at 3,000 ft

2611+12: 260° at 11 kt, +12 °C · 2824-01: 280° at 24 kt, –1 °C

781550: direction over 50 means 100+ kt: 280° at 115 kt, –50 °C (no sign above 24,000)

9900: light and variable, under 5 kt

None within 1,500 ft of station elevation for wind, 2,500 ft for temperature.

True or magnetic?

METAR, TAF and FB winds: **true**.

ATIS, ASOS/AWOS voice, the tower, and PIREP /WV: **magnetic**.

PIREP order

Station · UA, or UUA if urgent · /OV where · /TM time UTC · /FL altitude MSL · /TP type · /SK sky · /WX weather · /TA temp · /WV wind · /TB turbulence · /IC icing · /RM remarks

Sources: FAA-H-8083-28B Ch. 24 (Fig. 24-1, Tables 24-3, 24-4, Fig. 24-5), FAA-H-8083-28B Ch. 27 (Tables 27-1, 27-3; FB and TAF sections), AIM 7-1-10, AIM 7-1-18 (PIREP element code chart), AIM 7-1-28 and 7-1-29 (NWS TAFs exclude BECMG), AIM 4-2-10. Original card, not an FAA document. Study aid, not for navigation.